

Task 2 : Data Cleaning

```
import numpy as np

# Make a copy of dataset
df1 = df.copy()

# Convert 'totalcharges' to numeric
df1["TotalCharges"] = pd.to_numeric(df1["TotalCharges"], errors="coerce")

# Remove duplicates
df1 = df1.drop_duplicates()

# Clean churn column
df1['Churn'] = df1['Churn'].astype(str).str.strip().str.lower().replace({'yes': 'Yes', 'no': 'No'})

# Save cleaned dataset
df1.to_csv("cleaned_dataset.csv", index=False)
print("✅ Data cleaning complete! Saved as 'cleaned_dataset.csv'")

✅ Data cleaning complete! Saved as 'cleaned_dataset.csv'
```